

SECURITY ASSESSMENT

Grand Marina Hotel Water Monitoring System

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EXECUTIVE SUMMARY

Grand Marina Hotel should deploy TLS encryption for its water monitoring system because testing showed that it protects sensor data without significantly affecting performance. The security tests successfully blocked the tested unauthorized access, while TLS added only 0.23% average latency and the system handled up to 100 messages per second with 100% success and no errors. Based on these results, the security benefits outweigh the very small performance impact.

SECURITY RISKS ADDRESSED

Before Security:

The water monitoring data is currently sent across the hotel network without encryption. During testing, we were able to monitor the sensor traffic and read the information being transmitted, including the device location, water pressure, flow rate, and other sensor readings.

This means that someone who gains access to the hotel network could potentially observe the hotel's water system activity without being detected. They could learn how the system is operating, identify unusual conditions, or use the information to target the hotel's water infrastructure. For a hotel serving more than 2,000 guests per

night, unauthorized access to this monitoring system could affect operations, guest experience, and the hotel's reputation.

After Security:

With encryption enabled, the sensor data is protected while it travels across the hotel network. During testing, an unauthorized subscriber was unable to access the sensor messages, so the readings could no longer be viewed as readable information by the tested eavesdropper.

The system also verifies the broker's digital certificate before establishing a trusted connection. When an incorrect certificate was tested, the connection was rejected. This helps prevent unauthorized systems from pretending to be the trusted monitoring system and protects the hotel's water monitoring data from unauthorized access.

PERFORMANCE IMPACT

What We Measured	Before Security	After Security	Verdict
How fast readings arrive	51.95 ms	52.07 ms	Only 0.23% slower
How many sensors at once	100 messages/sec	100 messages/sec	Handles all tested loads

TLS increased the average time for a reading by only 0.12 milliseconds, which is a very small change that would not be noticeable during normal hotel operations. The system also successfully handled up to 100 messages per second with security enabled, with no errors, showing that encryption did not reduce performance under any of the tested loads.

TESTING EVIDENCE

We ran four security tests on the water monitoring system:

1. Can outsiders see sensor data?

Without security, we were able to monitor the sensor traffic and read the transmitted data. After encryption was enabled, the tested unauthorized subscriber was unable to access the sensor messages.

2. Does the system block unauthorized connections?

The system successfully connected when the correct trusted certificate was used. When we tested an incorrect certificate, the connection was rejected. This showed that the system can prevent connections when the presented certificate is not trusted.

3. Does security slow things down?

Average latency increased from 51.95 milliseconds without encryption to 52.07 milliseconds with encryption. This is only a 0.23% increase, or 0.12 milliseconds, making the difference effectively unnoticeable during normal operations.

4. Can the system handle a crisis?

The system successfully handled all tested load levels from 10 to 100 messages per second, both with and without encryption. Every test achieved 100% success with zero errors, including the highest tested rate of 100 messages per second.

RECOMMENDATION

Grand Marina Hotel should deploy TLS encryption for the water monitoring system. Our testing showed that encryption successfully protected the tested sensor traffic while adding only 0.23% average latency, and the system continued to achieve 100% success with zero errors at every tested load level up to 100 messages per second.

This is particularly important because a security incident can create significant financial and operational consequences. [IBM's 2025 Cost of a Data Breach Report](#) found a global average breach cost of \$4.44 million across more than 600 studied organizations. Based on the security improvement demonstrated in testing and the very small measured performance impact, deploying encryption is justified.

Source: IBM, *Cost of a Data Breach Report 2025* —[IBM Official Report Page](#)

NEXT STEPS

1. **Enable encrypted connections for all three water monitoring sensors.** Move the hotel water monitoring system from unprotected communication to encrypted communication so that sensor data cannot be easily read while traveling across the hotel network.
2. **Restrict who can connect to the monitoring system.** Give only authorized sensors and monitoring systems the credentials needed to connect, and block devices that do not have valid credentials.
3. **Monitor and document the secured system.** Set up monitoring for failed connection attempts and unusual activity, and keep records of the security configuration and test results. Re-test the system after deployment to confirm that encryption and access controls continue to work as expected.